Anchored Logarithmic Primitives on Translated Fibers: Exact Tauberian Defects, Uniformity Regimes, and the Literal Reassembly Gate

Liang Wang
School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

Let $a_1, \ldots, a_N \in \mathbb{C}$, let $r \geq 0$ be real translation, and put

$$S_N = \sum_{n \leq N} a_n, \qquad L_{r,N} = \sum_{n \leq N} \frac{a_n}{r+n}, \qquad L_{r,0} = 0.$$

The logarithmic primitive on a translated interval has an exact Tauberian inverse that differs from the unshifted one:

$$\boxed{S_N = (r+N)L_{r,N} - \sum_{m<N} L_{r,m}, \qquad \frac{S_N}{N} = \left(1 + \frac{r}{N}\right) L_{r,N} - \frac{1}{N} \sum_{m<N} L_{r,m}.}$$

Thus ordinary cancellation is equivalent to vanishing of an *anchored backward Cesàro defect*. The extra term $(r/N)L_{r,N}$ is not an error that may be dropped after translating a physical fiber.

We prove the exact terminal criterion: once the unshifted signed defect is $o(1)$, ordinary cancellation holds if and only if $(r/N)L_{r,N} = o(1)$. Thus $r = 0$ needs no terminal condition; for $0 < r = o(N)$ the exact scale is $L_{r,N} = o(N/r)$, with boundedness only a convenient stronger hypothesis; for $r \asymp N$ one needs $L_{r,N} = o(1)$; and for $r \gg N$ one needs $L_{r,N} = o(N/r)$. The constant sequence gives the exact obstruction:

$$L_{r,N} = \sum_{n \leq N} \frac{1}{r+n} \sim \frac{N}{r} = o(1) \quad (r/N \to \infty), \qquad S_N/N = 1.$$

Hence even an unnormalized $o(1)$ logarithmic statement can be useless on a short translated fiber.

For a physical packet

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} S_{\theta,X}(N_{\theta,X}) + E_{\mathrm{ret},X},$$

we obtain the exact signed aggregate formula

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} N_{\theta,X} \mathcal{T}_{\rho_{\theta,X}, N_{\theta,X}}(L_{\theta,X}) + E_{\mathrm{ret},X}.$$

This identifies the quantitative certificate that a logarithmic theorem must return after every literal origin, prefix, outer coefficient, mask, and content term has been restored. The identities and counterexamples are L0; attachment to a complete native packet is an L1 certificate. No uniform estimate for the growing fixed-h_0 affine Möbius family is proved, so there is no new L2 arithmetic saving, parity breakthrough, or prime-pair consequence.

Keywords: logarithmic average; translated interval; Tauberian theorem; Cesàro defect; affine Möbius correlation; fixed shift; outer reassembly.

Literal-frame convention. Every physical statement retains one prescribed nonzero integer h_0, the actual growing affine forms and interval origins, the literal masks and weights, every native key and polarization, and one global post-aggregation normalization. Fixed-form, averaged-shift, almost-all-scale, or translated-model estimates keep those qualifiers.

1 Introduction

Logarithmic correlation theorems naturally weight an arithmetic variable m by $1/m$. A physical affine fiber, however, is often written after translation as

$$m = r + n, \qquad 1 \le n \le N.$$

The inherited weight is $1/(r+n)$, not $1/n$. Replacing it by $1/n$ changes both the primitive and its Tauberian inverse.

TPC-87 isolated the exact signed Cesàro defect for the unshifted primitive and warned that a translated reciprocal weight must be transferred rather than relabelled [4]. The present paper computes that transfer exactly. The new term is elementary, but its scale is decisive when the physical origin is comparable to or larger than the fiber length.

The distinction is also required when comparing the literal packet with logarithmically averaged Chowla and Elliott theorems. Tao's two-point theorem has fixed affine coefficients and a logarithmic window [2]. Averaged-shift and almost-all-scale results have different quantifiers [1, 3]. None may be attached to a polynomially growing, prescribed fixed-h_0 family without checking coefficients, origins, prefixes, masks, and the outer sum.

1.1 Main contributions

We prove:

(i) the exact anchored Abel identity and its necessary-and-sufficient signed defect;

(ii) the exact necessary-and-sufficient terminal condition and its consequences in the regimes $r = o(N)$, $r \asymp N$, and $r \gg N$;

(iii) exact counterexamples showing that unnormalized logarithmic smallness and per-fiber qualitative cancellation do not survive translation or growing outer mass;

(iv) a prefix-uniform positive criterion for genuinely short translated fibers;

(v) the signed and absolute aggregate certificates for the complete literal affine packet; and

(vi) a separate determinant and physical-loss ledger that prevents a Tauberian normalization change from being counted as arithmetic cancellation.

1.2 What changes in the TPC route

The correct fiber-level datum is

$$\mathcal{T}_{r,N}(L) = \left(1 + \frac{r}{N}\right) L_N - \frac{1}{N} \sum_{m<N} L_m,$$

not the unshifted defect alone. This refines the Tauberian branch of the TPC-92 roadmap [5]. It does not estimate the datum for the actual Möbius coefficient. That remains an arithmetic problem after the literal low-frequency export of TPC-93 [6].

2 The anchored Abel identity

Definition 2.1 (Translated logarithmic primitive). For $r \ge 0$, define

$$S_N = \sum_{n=1}^{N} a_n, \qquad L_{r,N} = \sum_{n=1}^{N} \frac{a_n}{r+n}, \qquad L_{r,0} = 0.$$

Theorem 2.2 (Exact anchored Tauberian identity)**.** *For every complex sequence, every real $r \geq 0$, and every $N \geq 1$,*

$$S_N = (r+N)L_{r,N} - \sum_{m=0}^{N-1} L_{r,m}, \tag{1}$$

$$\frac{S_N}{N} = \left(1 + \frac{r}{N}\right) L_{r,N} - \frac{1}{N} \sum_{m=0}^{N-1} L_{r,m}. \tag{2}$$

Proof. Since

$$a_n = (r+n)(L_{r,n} - L_{r,n-1}),$$

summation and a shift of index give

$$\begin{aligned} S_N &= \sum_{n=1}^{N} (r+n)L_{r,n} - \sum_{m=0}^{N-1} (r+m+1)L_{r,m} \\ &= (r+N)L_{r,N} - \sum_{m=0}^{N-1} L_{r,m}, \end{aligned}$$

because $L_{r,0} = 0$. Division by N proves the second identity. □

Definition 2.3 (Anchored backward defect)*.* For a finite sequence $L = (L_m)_{0 \leq m \leq N}$ with $L_0 = 0$, put

$$\mathcal{T}_{r,N}(L) := \left(1 + \frac{r}{N}\right) L_N - \frac{1}{N} \sum_{m=0}^{N-1} L_m. \tag{3}$$

The usual signed backward Cesàro defect is

$$\mathcal{T}_{0,N}(L) = L_N - \frac{1}{N} \sum_{m<N} L_m.$$

Corollary 2.4 (Exact decomposition of the translation cost)**.**

$$\boxed{\mathcal{T}_{r,N}(L) = \mathcal{T}_{0,N}(L) + \frac{r}{N} L_N.} \tag{4}$$

Consequently $S_N/N \to 0$ along any family of triples (r, N, a) if and only if $\mathcal{T}_{r,N}(L_r) \to 0$.

Proof. Subtract the two definitions and use equation (2). □

Remark 2.5 (The anchor is part of the main term)*.* Equation (4) is exact. There is no reason for $(r/N)L_N$ to be small merely because $L_N = o(1)$. The relevant terminal scale is N/r when $r \gg N$.

Proposition 2.6 (Absolute oscillation bound)**.** *Define*

$$\mathcal{A}_N(L) = \frac{1}{N} \sum_{m=0}^{N-1} |L_N - L_m|.$$

Then

$$\left| \frac{S_N}{N} \right| \leq \mathcal{A}_N(L_r) + \frac{r}{N} |L_{r,N}|. \tag{5}$$

Proof. Write the unshifted defect as

$$\mathcal{T}_{0,N}(L) = \frac{1}{N} \sum_{m<N} (L_N - L_m)$$

and apply the triangle inequality in (4). □

3 Sharp translation regimes

Theorem 3.1 (Exact terminal criterion and regime corollaries)**.** *Consider a family indexed by X, with anchors r_X, lengths $N_X \to \infty$, primitives L_X, and ordinary sums S_X. Assume*

$$\mathcal{T}_{0,N_X}(L_X) = o(1).$$

Then

$$\frac{S_X}{N_X} = o(1) \quad \Longleftrightarrow \quad \frac{r_X}{N_X} L_X(N_X) = o(1). \tag{6}$$

Consequently:

(a) *If $r_X = 0$, no terminal condition is needed.*

(b) *If $0 < r_X = o(N_X)$, the exact terminal scale is $L_X(N_X) = o(N_X/r_X)$; the stronger bound $L_X(N_X) = O(1)$ is a convenient sufficient condition.*

(c) *If $r_X \asymp N_X$, then the two conditions*

$$\mathcal{T}_{0,N_X}(L_X) = o(1), \qquad L_X(N_X) = o(1)$$

imply $S_X/N_X = o(1)$.

(d) *If $r_X/N_X \to \infty$, then*

$$\mathcal{T}_{0,N_X}(L_X) = o(1), \qquad L_X(N_X) = o(N_X/r_X)$$

imply $S_X/N_X = o(1)$.

The terminal condition in (6) is necessary and sufficient without additional structure.

Proof. The identity

$$\mathcal{T}_{r_X,N_X}(L_X) = \mathcal{T}_{0,N_X}(L_X) + \frac{r_X}{N_X} L_X(N_X)$$

and $S_X/N_X = \mathcal{T}_{r_X,N_X}(L_X)$ prove (6). The regime statements are immediate. The examples in examples 4.1 and 4.2 show that the anchor term cannot be removed. □

For a positive comparison scale define the translated harmonic mass

$$\mathcal{H}_{r,N} := \sum_{n=1}^{N} \frac{1}{r+n}. \tag{7}$$

Lemma 3.2 (Short translated harmonic mass)**.** *For $r \geq N \geq 1$,*

$$\frac{N}{r+N} \leq \mathcal{H}_{r,N} \leq \frac{N}{r}, \qquad \left(1 + \frac{r}{N}\right) \mathcal{H}_{r,N} \leq 2. \tag{8}$$

Proof. Each denominator lies between $r+1$ and $r+N$. The displayed bounds follow by comparison; the last uses $N/r \leq 1$. □

Theorem 3.3 (Prefix-uniform criterion for short translated fibers)**.** *Assume $r \geq N$ and*

$$\max_{1 \leq m \leq N} \frac{|L_{r,m}|}{\mathcal{H}_{r,m}} \leq \varepsilon. \tag{9}$$

Then

$$\left| \frac{S_N}{N} \right| \leq 3\varepsilon. \tag{10}$$

Proof. By theorem 2.2,

$$\begin{aligned}\left|\frac{S_N}{N}\right| &\le \left(1+\frac{r}{N}\right)|L_{r,N}| + \frac{1}{N}\sum_{m<N}|L_{r,m}| \\ &\le 2\varepsilon + \frac{\varepsilon}{N}\sum_{m<N}\mathcal{H}_{r,m}.\end{aligned}$$

Since $\mathcal{H}_{r,m} \le m/r \le 1$, the last average is at most ε. □

Remark 3.4 (The quantifier is strong). The hypothesis is uniform in every prefix and normalized by its actual translated harmonic mass. A statement only at the terminal endpoint, or one with an $o(1)$ error not measured relative to $\mathcal{H}_{r,m}$, does not imply (9). Moreover the case $m = 1$ reads $|a_1| \le \varepsilon$. For a unit-modulus arithmetic sequence this cannot hold with $\varepsilon = o(1)$. A realistic application must peel a finite initial segment, return it exactly to the outer sum, and impose the relative estimate only on the remaining prefixes. The theorem is a strong sufficient certificate, not a claim that literal Möbius fibers already satisfy it.

Corollary 3.5 (Cutoff-prefix certificate). *Assume $r \ge N$, fix $1 \le M \le N$, and suppose*

$$\sup_{M\le m\le N}\frac{|L_{r,m}|}{\mathcal{H}_{r,m}} \le \varepsilon.$$

Then

$$\left|\frac{S_N}{N}\right| \le 3\varepsilon + \frac{1}{N}\sum_{m=0}^{M-1}|L_{r,m}|. \tag{11}$$

Proof. Use equation (2). The terminal term is at most

$$\left(1+\frac{r}{N}\right)\varepsilon\mathcal{H}_{r,N} \le \varepsilon\left(1+\frac{N}{r}\right) \le 2\varepsilon.$$

The controlled part of the backward average is at most

$$\frac{\varepsilon}{N}\sum_{m=M}^{N-1}\mathcal{H}_{r,m} \le \varepsilon,$$

and the omitted prefixes give the last term in (11). □

Proposition 3.6 (Smooth terminal bound). *Suppose, for every $0 \le m \le N$,*

$$L_{r,m} = \ell_{r,m} + e_{r,m},$$

where

$$\mathcal{T}_{0,N}(\ell) = 0.$$

Then

$$\left|\frac{S_N}{N}\right| \le \frac{r}{N}|\ell_{r,N}| + \left(1+\frac{r}{N}\right)|e_{r,N}| + \frac{1}{N}\sum_{m<N}|e_{r,m}|.$$

Thus a model primitive transfers only after its anchor term and the full prefix error have both been controlled.

Proof. Insert $L = \ell + e$ into (2) and use the assumed cancellation of the unshifted model defect. □

4 Exact counterexamples and sharpness

Example 4.1 (Vanishing primitive, nonvanishing ordinary mean)**.** *Let $a_n = 1$, and let $r_N/N \to \infty$. Then*

$$\frac{S_N}{N} = 1, \qquad L_{r_N,N} = \mathcal{H}_{r_N,N} = \frac{N}{r_N}\{1 + o(1)\} = o(1).$$

Moreover

$$\mathcal{T}_{0,N}(L_{r_N,\cdot}) = o(1), \qquad \frac{r_N}{N} L_{r_N,N} = 1 + o(1).$$

Thus the entire ordinary mean is carried by the anchor term.

Proof. The harmonic estimate follows uniformly from

$$\frac{N}{r_N + N} \le \mathcal{H}_{r_N,N} \le \frac{N}{r_N}.$$

The anchored identity gives the last assertion; alternatively, $L_{r_N,m} = m/r_N + O(m^2/r_N^2)$ uniformly for $m \le N$. □

Example 4.2 (Tuned terminal scale)**.** *Let $r \ge N$ and let $a_n = \xi$ be constant. Then*

$$\frac{S_N}{N} = \xi, \qquad L_{r,N} = \xi \mathcal{H}_{r,N}.$$

Choosing $\xi = \delta r/N$ gives

$$L_{r,N} = \delta\{1 + O(N/r)\}, \qquad S_N/N = \delta r/N.$$

Hence a terminal primitive of order δ can amplify by r/N, and the scale N/r in theorem 3.1 cannot be enlarged uniformly.

Example 4.3 (Per-fiber $o(1)$ does not aggregate)**.** *For each integer M, take M fibers with $r_\theta = 0$, $N_\theta = 1$, $c_\theta = 1$, and $a_{\theta,1} = 1/M$. Each fiber satisfies*

$$|\mathcal{T}_{0,1}(L_\theta)| = \frac{1}{M} = o(1),$$

but

$$\sum_{\theta=1}^{M} c_\theta N_\theta \mathcal{T}_{0,1}(L_\theta) = 1.$$

Thus a qualitative bound on every fiber is not an aggregate certificate when the outer coefficient mass grows.

Remark 4.4 (Scope of the counterexamples)*.* The examples are structure-free L0 obstructions. They do not give lower bounds for the literal affine Möbius packet. They prove that a proposed transfer theorem using only the displayed hypotheses is false.

Proposition 4.5 (No anchor-free black box)**.** *There is no function $\Phi(\varepsilon) \to 0$ such that, uniformly for all r, N and all $|a_n| \le 1$,*

$$|L_{r,N}| \le \varepsilon, \qquad |\mathcal{T}_{0,N}(L_r)| \le \varepsilon \quad \Longrightarrow \quad |S_N|/N \le \Phi(\varepsilon).$$

Proof. Use example 4.1 and choose $r_N/N \to \infty$ fast enough that both hypotheses tend to zero, while $S_N/N = 1$. □

5 Prefix-complete aggregate reassembly

Let Θ_X be a finite native-key set. For each $\theta \in \Theta_X$, let $\rho_{\theta,X} \geq 0$ be the literal origin, $N_{\theta,X} \geq 1$ the returned length, and

$$b_{\theta,X}(n), \qquad 1 \leq n \leq N_{\theta,X},$$

the complete local coefficient after retaining the affine Möbius factors, periodic factors, smooth weight, residue and coprimality masks, and the additive determinant phase when present. Define

$$S_{\theta,X}(N) := \sum_{n \leq N} b_{\theta,X}(n), \tag{12}$$

$$L^{(\rho)}_{\theta,X}(N) := \sum_{n \leq N} \frac{b_{\theta,X}(n)}{\rho_{\theta,X} + n}. \tag{13}$$

The symbol $\rho_{\theta,X}$ is an interval origin; it is unrelated to the Fourier frequency r used in the TPC-93 low window.

Theorem 5.1 (Exact anchored packet identity)**.** *Suppose the literal raw packet has the exact reassembly*

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} S_{\theta,X}(N_{\theta,X}) + E_{\mathrm{ret},X}. \tag{14}$$

Then

$$\boxed{Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} N_{\theta,X} \mathcal{T}_{\rho_{\theta,X},N_{\theta,X}}(L^{(\rho)}_{\theta,X}) + E_{\mathrm{ret},X}.} \tag{15}$$

In particular,

$$\begin{aligned} |Z_X| \leq \sum_{\theta \in \Theta_X} & |c_{\theta,X}| N_{\theta,X} \left| \mathcal{T}_{\rho_{\theta,X},N_{\theta,X}}(L^{(\rho)}_{\theta,X}) \right| \\ & + |E_{\mathrm{ret},X}|. \end{aligned} \tag{16}$$

Proof. Apply theorem 2.2 to every finite fiber in (14). No limiting interchange is required. The inequality follows after the exact outer sum has been formed. □

Remark 5.2 (Literal TPC-93 convention)*.* In a TPC-93 application the native key must include the low Fourier frequency, polarization, exact-content child, masks, and interval component. The sign in the exact relation

$$Z_X = -\operatorname{Low}_X + \operatorname{Tail}_X,$$

together with the reconstruction multiplier and each frequency-dependent unit phase, stays in $c_{\theta,X}$ or $b_{\theta,X}$ according to its literal location. The returned remainder is then

$$E_{\mathrm{ret},X} = \operatorname{Tail}_X,$$

not an invented content error. The complete source–child inverse and multiplicity reconstruction established in TPC-93 supplies this L1 attachment [6].

Corollary 5.3 (Quantitative zero-mode certificate)**.** *At the critical physical scales, if*

$$\begin{aligned} \sum_{\theta \in \Theta_X} |c_{\theta,X}| N_{\theta,X} & \left| \mathcal{T}_{\rho_{\theta,X},N_{\theta,X}}(L^{(\rho)}_{\theta,X}) \right| + |E_{\mathrm{ret},X}| \\ & \leq X^{-\eta z + o(1)} Q_X^2, \end{aligned} \tag{17}$$

then

$$|Z_X| \leq X^{-\eta z + o(1)} Q_X^2.$$

Proof. Use (16). □

Remark 5.4 (Signed aggregate is the minimal target). The absolute certificate (17) is sufficient, not necessary. The exact minimal object is the signed outer sum in (15). Taking absolute values before native outer reassembly may lose a fixed power. Any argument exploiting outer cancellation must still retain every native coefficient and origin.

Remark 5.5 (Bounded fibers). A fiber with $N_{\theta,X} = O(1)$ has no internal asymptotic parameter. It remains in (15) exactly and must be controlled by its coefficient mass or by cancellation after outer reassembly. Deleting it because a long-fiber theorem does not apply changes Z_X.

Proposition 5.6 (Coefficient-mass promotion rule)**.** *Suppose a subset $\Theta_X^\star$ satisfies*

$$\left|\mathcal{T}_{\rho_{\theta,X},N_{\theta,X}}(L_{\theta,X}^{(\rho)})\right| \le \varepsilon_X \quad (\theta \in \Theta_X^\star).$$

Its absolute contribution is at most

$$\varepsilon_X \sum_{\theta\in\Theta_X^\star} |c_{\theta,X}| N_{\theta,X}.$$

Therefore a uniform $o(1)$ defect is useful only relative to the actual weighted outer mass.

Proof. This is the corresponding part of (16). □

6 Quantifier audit for logarithmic correlation theorems

The two-point logarithmically averaged Chowla theorem controls fixed affine forms on a multiplicative interval with the reciprocal weight of the original arithmetic variable [2]. The averaged Chowla theorem averages the shifts [1]. Structural results at almost all scales retain an almost-all qualifier [3]. These are major arithmetic inputs, but their quantifiers are not interchangeable with the literal TPC packet.

Proposition 6.1 (Literal promotion checklist)**.** *A logarithmic theorem can be promoted to a certificate for* (14) *only after verifying:*

(i) *the same prescribed h_0 and the actual slopes and intercepts, including their growth with X;*

(ii) *the original arithmetic variable and therefore the true anchor $\rho_{\theta,X}$;*

(iii) *uniformity in every prefix required by partial summation or dyadic recursion;*

(iv) *all interval origins, masks, smooth weights, periodic factors, polarizations, and native outer keys;*

(v) *an exact return of short, resonant, content, and endpoint sectors; and*

(vi) *a quantitative aggregate rate meeting* (17)*, or a direct bound for the signed sum* (15)*.*

Proof. Items (i)–(v) identify the sequence and frame in (13). Item (vi) is precisely the hypothesis needed by theorem 5.1. Omitting the anchor contradicts proposition 4.5; omitting outer mass contradicts example 4.3. □

Remark 6.2 (Fixed forms versus growing forms). Tao's theorem is stated for fixed a_1, a_2, b_1, b_2 with nonzero determinant. The literal columns have slopes, intercepts, origins, and masks depending on X, while preserving one fixed nonzero determinant h_0 after the exact content reduction. No uniformity in that entire growing family is inferred here.

Remark 6.3 (Averaged shifts do not select h_0). An estimate averaged over many shifts can show that most shifts are good without identifying the one prescribed physical shift. It remains at its averaged level until a literal selection theorem is proved.

Remark 6.4 (Logarithmic normalization). A normalized bound $L_{r,N} = o(\mathcal{H}_{r,N})$ has different strength in different regimes. For $r \ge N$, a prefix-uniform version is enough by theorem 3.3. For $r = 0$, $\mathcal{H}_{0,N} \asymp \log N$, and qualitative $o(\log N)$ cancellation alone does not control the unshifted signed defect; this is the sharp gap of TPC-87.

7 The literal fixed-shift ledger

At the critical row retain

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}, \qquad N_{0,X} = J_X Q_X^2.$$

Write the determinant-energy lower loss and zero-mode gain as

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2, \qquad |Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2.$$

The determinant compatibility remains

$$\lambda_D \leq 2\eta_Z. \tag{18}$$

Equality is usable but leaves no reserve.

Definition 7.1 (Physical loss ledger). Let Λ_{phys} be the sum of fixed-power losses actually incurred by coefficient growth, localization, translation, prefix return, masks, short-fiber treatment, content completion, outer reassembly, dictionary transfer, and downstream calibrated drift, after removing every loss already represented by $\lambda_D - 2\eta_Z$. The physical route continues only if

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}.} \tag{19}$$

Proposition 7.2 (No saving from an identity)**.** *The passage from* (14) *to* (15) *spends no fixed power and supplies no positive* η_Z*. A positive* η_Z *is certified only by a quantitative estimate for the actual signed aggregate or a sufficient absolute bound such as* (17)*.*

Proof. Equation (15) is equality for the same coefficient. Changing coordinates cannot by itself reduce its absolute value. □

Remark 7.3 (Independent accounts). The compatibility (18) compares two statistics of the same post-bin coefficient. The strict inequality (19) records losses in separate physical maps. They are not added into one budget, and the same translation or localization loss is never charged twice.

Proposition 7.4 (Anchored branch stop test)**.** *A rigorous parameter-consistent lower bound showing that every allowed logarithmic input in a specified class has*

$$\left| \sum_\theta c_{\theta,X} N_{\theta,X} \mathcal{T}_{\rho_{\theta,X},N_{\theta,X}}(L_{\theta,X}^{(\rho)}) \right| \geq X^{-\eta_Z+\delta+o(1)} Q_X^2$$

for some fixed $\delta > 0$*, together with*

$$|E_{\text{ret},X}| = o\left(\left| \sum_\theta c_{\theta,X} N_{\theta,X} \mathcal{T}_{\rho_{\theta,X},N_{\theta,X}}(L_{\theta,X}^{(\rho)}) \right| \right),$$

kills that transfer class at the claimed target. Equivalently, one may prove the corresponding lower bound directly for the complete Z_X*. A lower bound merely of the same* $X^{-\eta_Z+o(1)}Q_X^2$ *scale does not contradict an upper bound at that scale. Likewise, an unavoidable translation loss with* $\Lambda_{\text{phys}} \geq 1/400$ *stops the assembled physical route. Failure of a currently available theorem to supply uniformity is not such a lower bound.*

Proof. The fixed gap X^δ, and the exclusion of cancellation by the returned remainder, force the complete coefficient above the claimed target. The physical-loss assertion is the strict endpoint rule. The last sentence distinguishes an absent input from an impossibility theorem. □

8 Claim boundary and next gate

8.1 Exact results

The following are unconditional finite L0 statements:

(i) the anchored Abel identity (2);

(ii) the decomposition $\mathcal{T}_{r,N} = \mathcal{T}_{0,N} + (r/N)L_N$;

(iii) the regime criteria and the prefix-uniform short-fiber theorem;

(iv) the sharp counterexamples in section 4; and

(v) the aggregate identity (15), conditional only on being supplied the exact finite reassembly (14).

Verifying that the θ-family in (14) is the complete raw hard packet of the TPC-93 low-frequency export, with no missing native sector, is an L1 provenance certificate. It does not become L2 merely because the identity is attached.

8.2 Unproved arithmetic

This paper does not prove:

(a) a quantitative anchored-defect estimate for the actual growing affine Möbius fibers;

(b) uniformity of a fixed-form logarithmic theorem in the growing slopes, intercepts, origins, masks, or prefixes;

(c) selection of the prescribed h_0 from an averaged-shift theorem;

(d) control of the complete multiplier-weighted low-frequency window;

(e) a lower bound for M_X or D_X;

(f) TPC-18 full-block reassembly or primitive determinant dispersion;

(g) a fixed-shift Hardy–Littlewood asymptotic, parity breakthrough, prime-pair lower bound, or the twin-prime conjecture.

8.3 Audited conclusion

Theorem 8.1 (Anchored transfer conclusion)**.** *For every translated physical fiber, ordinary cancellation is equivalent to cancellation of the anchored defect*

$$\mathcal{T}_{r,N}(L) = \left(1 + \frac{r}{N}\right) L_N - \frac{1}{N} \sum_{m<N} L_m.$$

For a complete native packet the exact target is the signed aggregate (15)*; the absolute expression* (17) *is a sufficient quantitative certificate. Neither an anchor-free defect, a terminal logarithmic bound without prefix control, nor per-fiber* $o(1)$ *without weighted outer mass is a valid substitute.*

Proof. The fiber statement is theorem 2.2. Summing it with the literal coefficients gives theorem 5.1. The three forbidden substitutions are ruled out respectively by proposition 4.5, remark 3.4, and example 4.3. □

Next gate. Insert the literal origins and prefixes exported by TPC-93 and classified by the resonance/short-fiber analysis of TPC-94 [7]. The next positive theorem must estimate the resulting signed aggregate at scale $X^{-\eta z+o(1)}Q_X^2$. A negative theorem may instead prove that a specified logarithmic class cannot meet the anchored certificate. Either result stays at its proved scope.

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